Model Complexity, Regularisation, and Coefficient Stability in Binary and Multinomial Logistic Regression

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***Abstract*—In this research I look at how model complexity and regularisation interact in a case with limited dataset size and how this affects both predictive performance and estimator stability. Using a publicly available heart disease dataset, I compare binary logistic regression and multinomial logistic regression under maximum likelihood estimation (MLE) and ridge regularisation. The multinomial models provide greater flexibility by modeling disease severity levels, however they substantially increase the number of parameters and reduce the number of samples in each class. I evaluate models using cross-validated log-loss and assess estimator stability using a leave-one-out jackknife procedure. I show that the multinomial MLE exhibits large coefficients and jackknife variability for rare outcome classes, indicating finite sample instability. Coefficient shrinking reduces the instability and improves predictive performance, especially for the multinomial model. The simpler binary model achieves lower cross-validated log-loss overall. The findings illustrate how a more felxible model increases variance in cases where data is limited and how shrinkage mitigates this effect. The study highlights the practical trade-off between model complexity and stability.**

I. Introduction

Logistic regression is a widely used model for classification. In many cases analysts are faced with a fundamental modeling choice: whether to simplify the outcome of the data to a binary scale or keep a flexible, multinomial model that can predict several categories. While increased flexibility allows for richer modeling of the response variable, it also increases the number of parameters and may reduce estimator stability.

This trade-off between model complexity and statistical stability is particularly relevant when data are limited or when some outcome categories are rare. In such settings, maximum likelihood estimation (MLE) may yield large coefficient magnitudes, unstable standard errors, and poor out-of-sample performance. Regularisation techniques such as ridge regression are commonly used to mitigate these issues by shrinking coefficients toward zero, reducing variance at the cost of introducing bias. Understanding when this bias–variance trade-off becomes beneficial is a central question in statistics.

In this paper, I study this interaction between model complexity and regularisation using a heart disease dataset. The dataset contains clinical predictors and a categorical outcome variable indicating disease presence and severity. I am focusing on two modeling strategies:

- A binary logistic regression model, with disease severity simplified into a binary outcome (disease vs. no disease).
- A multinomial logistic regression model, where the original disease severity levels are used as a categorical outcome.

These two different approaches differ heavily in parameter space and amount of observations per outcome class. The multinomial model uses a lot more parameters which can increase variance and reduce stability, especially for less represented severity classes.

I compare four estimation approaches:

- Binary logistic regression estimated via MLE
- Binary logistic regression with ridge regularisation
- Multinomial logistic regression estimated via MLE
- Multinomial logistic regression with ridge regularisation

Model performance is evaluated using cross-validated log-loss. Estimators stability is measured via a leave-one-out jackknife procedure. This approach showcases how instability can change due to model complexity and how shrinkage can affect this phenomenon.

Research question:

- How do model complexity and ridge regularisation interact in finite samples, and how does this interaction affect both prediction accuracy and estimator stability?

Rather than focusing on finding relevant features for prediction, this paper examines statistical properties of estimation under different levels of flexibility. The results provide insight into when additional model complexity is beneficial and when it leads to undesirable variance and how well that can be mitigated by shrinkage.

II. Dataset and Modeling Format

A. Dataset Description

I use a public heart disease dataset from the UCI Machine Learning Repository [1]. It contains clinical and demographic information from four regions. After preprocessing and removal of incomplete records, the dataset consists of $n = 740$ observations. Each observation corresponds to one patient and includes a set of clinical and demographic predictors along with a categorical outcome variable indicating

heart disease severity.

The predictor variables can be grouped into the following categories:

- **Demographic variables:** age, sex.
- **Clinical measurements:** resting blood pressure, serum cholesterol, maximum heart rate achieved.
- **Diagnostic indicators:** chest pain type, fasting blood sugar indicator, resting electrocardiographic results, exercise-induced angina, ST depression, slope of peak exercise ST segment.
- **Cardiac imaging indicators:** number of major vessels observed via fluoroscopy, thalassemia status.

TABLE I
SUMMARY STATISTICS FOR CONTINUOUS PREDICTORS

Variable	Mean	Std	Min	Max
age	53.097	9.408	28.0	77.0
sex	0.765	0.424	0.0	1.0
cp	3.227	0.939	1.0	4.0
trestbps	132.754	18.581	0.0	200.0
chol	220.136	93.615	0.0	603.0
fbs	0.150	0.357	0.0	1.0
restecg	0.635	0.840	0.0	2.0
thalach	138.745	25.846	60.0	202.0
exang	0.400	0.490	0.0	1.0
oldpeak	0.894	1.087	-1.0	6.2

The outcome variable Y takes values in $\{0, 1, 2, 3, 4\}$, where 0 indicates no heart disease and higher values correspond to increasing severity of a heart condition.

TABLE II
DATASET OVERVIEW AND OUTCOME DISTRIBUTION

Quantity	Value	Notes
Observations (n)	740	After preprocessing
Variables (p)	13	
Outcome levels (`num`)	5	$\{0, 1, 2, 3, 4\}$
Class	**Count**	**Percent (%)**
`0`	357	48.2
`1`	204	27.6
`2`	79	10.7
`3`	78	10.5
`4`	22	3.0

The two modeling approaches:

- A **binary format**, where the outcome is collapsed into $Y \in \{0, 1\}$ (no disease vs. presence of disease).
- A **multinomial format**, where all five severity levels are kept and modeled.

The class distribution is imbalanced, with substantially fewer observations in the higher severity categories. Table I presents summary statistics for the predictors, and Table II shows the frequency distribution of the outcome classes.

The class imbalance implies reduced sample size for rare categories. This influences estimator stability and plays a central role in the analysis of a model with limited sample size later in the paper.

B. Preprocessing

Prior to modelling, several preprocessing steps were applied. Some predictors were fully missing outside the Cleveland subset (namely `slope`, `ca`, and `thal`) thus they were removed to ensure consistency across all observations. This resulted in a final set of $p = 12$ predictors.

Continuous predictors were standardised using z-score scaling, i.e.,

$$X_{scaled} = \frac{X - \mu}{\sigma},$$

where μ and σ denote the sample mean and standard deviation, respectively. This transformation yields variables with approximately zero mean and a variance of one. Standardisation is especially important for ridge regularisation, as it ensures that the penalty term treats affects all coefficients similarly.

Categorical predictors were encoded using one-hot (dummy) encoding, with one category omitted per variable.

I have further removed the origin column from the dataset as some regiens had very few observations in certain severity classes. This was especially prevalent in the multinomial model where the combination of rare classes and underrepresentation of regional severity classes led to some coefficients to explode. I have decided to keep a large mixed dataset, that does not represent locational data in any feature and instead focuses on patient features.

C. Modeling Format

Y denotes the outcome variable indicating heart disease severity. Two alternative models are analysed that differ in outcome complexity.

a) Binary

: In the binary setting, the outcome is defined as

$$Y \in \{0, 1\} \tag{1}$$

Where $Y = 0$ signifies no heart disease and $Y = 1$ signifies presence of some heart condition. The conditional probability of heart disease based on predictors X is modeled using logistic regression. Therefor the model requires estimation of 14 regression coefficients (plus an intercept) $\beta \in \mathbb{R}^p$.

b) Multinomial

: In the multinomial setting, the outcome is modelled as

$$Y \in \{0, 1, 2, 3, 4\} \tag{2}$$

Here, each non-reference class has its own parameter vector $\beta_k \in \mathbb{R}^p$, substantially increasing parameter dimensionality, having four times more parameters compared to the binary model.

c) Implications for variance

: The multinomial model increases the number of parameters by a factor of four (number of non-reference classes). In finite samples, this reduces the effective sample size per parameter, particularly for underrepresented severity classes. As shown in Table II, the highest severity class contains only 22 observations (3.0% of the data), which is very limited for estimating its coefficient vector.

This leads to unstable estimates for the multinomial model which serves as the basis of the bias-variance trade-off analysis and evaluation of the effect of shrinkage in stabilising high dimensional low sample logistic models.

III. Methodology

A. Logistic Regression and Maximum Likelihood Estimation

a) Binary Logistic Regression

: In this model the conditional probability of disease is modeled as

$$\Pr(Y = 1 \mid X) = \frac{1}{1 + \exp(-X^\top \beta)} \tag{3}$$

$\beta \in \mathbb{R}^p$ is the parameter vector. Equivalently, the model can be written in log-odds form:

$$\log \frac{\Pr(Y = 1 \mid X)}{\Pr(Y = 0 \mid X)} = X^\top \beta. \tag{4}$$

This highlights that logistic regression models a linear relationship between predictors and the log-odds of the outcome.

b) Multinomial Logistic Regression

: The model specifies

$$\Pr(Y = k \mid X) = \frac{\exp(X^\top \beta_k)}{1 + \sum_{j=1}^{4} \exp(X^\top \beta_j)}, \quad k = 1, \ldots, 4, \tag{5}$$

where each class k has its own coefficient vector $\beta_k \in \mathbb{R}^p$. In contrast to the binary model, the multinomial model requires estimation of multiple parameter vectors, substantially increasing dimensionality.

c) Maximum Likelihood Estimation

: Let $\{(x_i, y_i)\}_{i=1}^n$ denote the observed data. For the binary model, the log-likelihood is

$$\ell(\beta) = \sum_{i=1}^{n} \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right], \tag{6}$$

where $p_i = \Pr(Y = 1 \mid x_i)$

For the multinomial model, the log-likelihood becomes

$$\ell(\beta_k) = \sum_{i=1}^{n} \sum_{k=0}^{4} \mathbf{1}(y_i = k) \log \Pr(Y = k \mid x_i), \tag{7}$$

The maximum likelihood estimator (MLE) is obtained by solving the first-order optimality conditions. Under standard regularity assumptions and sufficiently large sample size, the MLE is asymptotically normal: $\hat{\beta} \approx \mathcal{N}\big(\beta, I(\beta)^{-1}\big)$ where $I(\beta)$ is the Fisher information matrix. The inverse information matrix determines the variance of the estimator.

d) Finite sample instability

: In finite samples, particularly when the number of parameters is large relative to the effective sample size, the Fisher information matrix may become poorly conditioned. This leads to inflated variance of the estimated coefficients. In the case of care categories , the number of observations that contribute information to a coefficient estimation is small and as a result the information matrix for those parameters is weak, which in turn increases estimator variability.

In extreme cases, quasi-separation may occur, where a specific set of predictors almost perfectly predict a class membership. Under maximum likelihood estimation, this leads to very large coefficients and unstable inference.

Consequently, while MLE estimates well on asymptotic data, its finite-sample behaviour can deteriorate when model dimensionality increases or when outcome classes are imbalanced. This motivates the consideration of regularisation techniques to stabilise estimation.

B. Ridge Regularisation

a) Penalised log-likelihood

: Ridge regression modifies the estimation problem by adding an $\ell 2$ penalty to the log-likelihood. For the binary model, the penalised objective becomes

$$\ell_{\text{ridge}}(\beta)\ell(\beta)\lambda|\beta|_2^2, \tag{8}$$

where $\ell(\beta)$ is the log-likelihood and $|\beta|_2^2$ is the squared euclidean norm of the coefficient vector. The parameter $\lambda \geq 0$ controls the strength of the penalty, if $\lambda = 0$, the estimator becomes the standard MLE. For the multinomial model, the penalty is applied to all class-specific parameter vectors.

b) Bias-Variance trade-off

: The ridge penalty shrinks coefficient estimates toward zero. This introduces bias but reduces variance, making the model less sensitive to random noise in the training data. As a result, the model may generalise better to unseen data when the unregularised estimator exhibits high variance.

By shrinking all coefficients toward zero, ridge regularisation prevents extreme estimates and mitigates separation effects. Consequently, it is particularly beneficial in high-dimensional or imbalanced settings such as the multinomial model in this research.

C. Model Evaluation via Cross-Validation

To evaluate model performance and select regularisation strength, I used K-fold cross-validation, where $K = 5$. The dataset was partitioned into K approximately equal-sized folds. For each fold the model was trained on $K - 1$ folds and evaluated on the left out fold. Performance metrics were averaged across all folds to obtain an estimate of out-of-sample prediction error. Preprocessing steps like scaling were done on the evaluation folds in each iteration separately to avoid data leakage.

a) Model Evaluation using log-loss
: Model performance was assessed using cross-validated log-loss. For the binary model, the log-loss is defined as

$$\text{LogLoss} - \frac{1}{n}\sum_{i=1}^{n}\left[y_i \log p_i + (1-y_i)\log(1-p_i)\right], \quad (9)$$

For the multinomial model, the log-loss generalises to

$$\text{LogLoss} - \frac{1}{n}\sum_{i=1}^{n}\sum_{k=0}^{K-1}\mathbf{1}(y_i = k)\log \Pr(Y = k \mid x_i). \quad (10)$$

Log-loss measures the negative log-likelihood of predicted probabilities and penalises heavily when the model confidently guesses the wrong outcome. As logistic regression is a probabilistic model estimated via likelihood maximisation, log-loss is a good evaluation metric for it.

b) Hyperparameter selection
: For ridge-regularised models, the regularisation strength was selected via grid search over a range of values of the inverse penalty parameter $C = \frac{1}{\lambda}$. For each candidate value of C, K-fold cross-validation was performed and the mean validation log-loss was computed. The value that minimises the average cross-validated log-loss is selected as optimal penalty factor for the ridge regression. The final ridge model was then refitted on the full dataset using the selected C. Separate hyperparameter tuning was conducted for the binary and multinomial models.

By selecting the regularisation strength that minimises cross-validated log-loss, the procedure implicitly balances bias and variance. In settings where model dimensionality is high or outcome classes are imbalanced, this selection mechanism is expected to favour stronger regularisation to stabilise estimation. Why log-loss is appropriate (probabilistic model comparison)

D. Estimator Stability via Jackknife

Cross-validation assesses predictive performance, but does not evaluate estimator stability. I used leave-one-out jackknife to evaluate finite sample variability of the estimators.

a) Leave-one-out jackknife
: Let $\{(x_i, y_i)\}_{i=1}^{n}$ denote the observed sample and let $\hat{\beta}$ be the parameter estimate obtained from the full dataset. LOO-jackknife constructs n reduced datasets by removing one observation at a time. For each removal the model is refitted on the dataset excluding the removed observation. The result is n parameter estimates $\hat{\beta}_{(-i)}$, each of them calculated on a dataset missing a different observation.

b) Jackknife variance estimation
: The jackknife estimate of the variance for coefficient β_j is given by

$$\widehat{\text{Var}}_{\text{jack}}(\hat{\beta}_j)\frac{n-1}{n}\sum_{i=1}^{n}\left(\hat{\beta}_{j(-i)} - \bar{\beta}_j\right)^2, \quad (11)$$

where β_j is the mean of the leave-one-out estimates. The square root of this quantity provides a jackknife standard error.

c) Interpretation: The jackknife measures how sensitive the estimator is to the removal of a single observation. If a small difference in the dataset leads to large fluctuations in estimated coefficients, the jackknife variance will be large, showing finite-sample instability.

If a model has sufficient information and stable likelihood, removing a single observation should not have a big effect on parameter estimates. If a model is high dimensional with class imbalance and some underrepresented classes, the estimator may rely too much on specific observations. In this case, the leave-one-out estimates can vary considerably, and the jackknife standard-errors will be larger.

d) Connection to the modeling objective
: The jackknife analysis complements cross-validation by focusing on estimator variability rather than predictive accuracy. Large jackknife standard errors signal weak information and poor conditioning of the likelihood, which aligns with theoretical expectations derived from the Fisher information matrix. Comparing jackknife standard errors across the binary and multinomial models therefore provides direct empirical evidence of how increased model complexity affects finite-sample stability and again comparing them with their ridge penalised versions one can observe the effect of shrinkage on model stability.

IV. Results

A. Baseline MLE Comparison

To assess the effect of model complexity on maximum likelihood estimation, I first compare the unregularised binary and multinomial logistic regression models.

TABLE III
Baseline MLE comparison

Model	# Params	LogLik	AIC	BIC
Binary Logit (MLE)	14	-310.89	649.78	714.28
Multinomial Logit (MLE)	53	-688.38	1482.77	1726.92

a) Model fit
: Table III summarises log-likelihood values and information criteria for both models. Although the multinomial model is more flexible, its log-likelihood is substantially lower than that of the binary model. This reflects the increased difficulty of modeling five outcome categories rather than two. The multinomial model distributes probability mass across more classes, resulting in higher uncertainty and lower per-observation log-likelihood. Information criteria such as AIC and BIC penalise model dimensionality. Given the substantially larger number of parameters in the multinomial case, BIC in particular assigns a strong penalty due to complexity.

b) Coefficient magnitudes
: The estimated coefficients reveal substantial differences in scale between the two models. The binary model produces moderate coefficient magnitudes across predictors, suggesting stable estimation.

In contrast, the multinomial model exhibits very large coefficients for certain predictors, most most likely those associated

TABLE IV
SELECTED COEFFICIENT COMPARISON (MLE)

Predictor	Binary $\hat{\beta}$	Max Multinomial $\hat{\beta}_k$	Class
age	0.20	0.73	2
chol	-0.18	-0.57	2
thalach	-0.36	-0.59	2
oldpeak	0.68	1.41	3
sex_1.0	1.30	2.33	3
cp_4.0	1.43	2.40	3
cp_2.0	-0.42	-8.82	3
exang_1.0	1.01	1.43	2

with rare outcome classes. Some class-specific coefficients reach magnitudes far exceeding those observed in the binary model.

Large coefficient magnitudes correspond to extreme log-odds and imply deterministic class probabilities. This behaviour is consistent with quasi-separation, where limited observations in specific combinations of class and predictor make the likelihood push parameters toward extreme values.

c) Rare class behaviour *:* The highest severity class contains only 22 observations (3.0% of the dataset). In the multinomial model, a separate coefficient vector must be estimated for this class. The small effective sample size provides limited information, resulting in unstable parameter estimates and inflated magnitudes.

By contrast, in the binary model, all diseased cases are pooled together, increasing the effective sample size and stabilising estimation.

d) Convergence *:* Both models converged numerically under maximum likelihood estimation. However, the multinomial model required more iterations and exhibited larger gradient fluctuations near convergence, reflecting weaker curvature of the likelihood surface.

e) Summary: Overall, the baseline comparison indicates that the multinomial model does not improve fit relative to the binary approach. Despite its greater flexibility, the multinomial model yields a substantially lower log-likelihood per observation and higher log-loss. This reflects the increased difficulty of modelling five outcome categories rather than two.

At the same time, the multinomial model exhibits clear signs of finite-sample instability, particularly for rare outcome classes. Several class-specific coefficients attain extreme magnitudes, suggesting weak information and near-separation. In contrast, the binary model produces more moderate coefficient estimates and appears more stable under maximum likelihood estimation, consistent with its lower effective dimensionality. Present:

B. Cross-Validated Prediction Performance

To evaluate out-of-sample predictive performance, all models were assessed using K-fold cross-validated log-loss. Table V summarises the results for the unregularised and ridge-regularised models.

TABLE V
CROSS-VALIDATED LOG-LOSS COMPARISON

Model	Best C	CV Log-Loss
Binary MLE	–	0.4454
Binary Ridge	0.3162	0.4412
Multinomial MLE	–	1.0568
Multinomial Ridge	0.1	0.9995

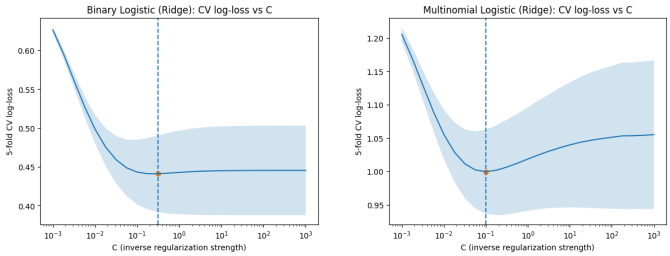


Fig. 1. Cross-validated log-loss as a function of C for binary (left) and multinomial (right) ridge models.

Figure 1 displays the cross-validated log-loss as a function of the regularisation parameter C for both ridge models.

a) Binary model *:* For the binary model, ridge regularisation yields only a small reduction in cross-validated log-loss compared to the unregularised MLE. The log-loss improved by a factor of approximately $\Delta = 0.0043$, indicating that the binary estimator already exhibits relatively low variance. The bias-variance curve in Figure 1 shows only minor variation in performance across a long range of C values, suggesting that the binary model is stable.

b) Multinomial model *:* In contrast, the multinomial approach benefits heavily from regularisation. Ridge reduces cross-validated log-loss by approximately $\Delta = 0.0573$, which is more than ten times the improvement observed in the binary case. The corresponding curve in Figure 1 exhibits a pronounced U-shape, indicating a clear bias-variance trade-off. Stronger regularisation (smaller C) initially reduces variance and improves performance, while excessive shrinkage eventually increases bias.

This behaviour is consistent with the high parameter dimensionality of the multinomial model and the presence of rare outcome classes, which increase estimator variance under maximum likelihood estimation.

c) Overall comparison *:* Despite the bigger improvement from ridge regularisation in the multinomial model, the binary model still achieves lower cross-validated log-loss. This indicates that, in this dataset, collapsing severity levels into a binary outcome leads to superior predictive performance. The results suggest that the additional flexibility of the multinomial model does not translate into better generalisation in this finite-sample case.

These findings support the hypothesis that increased model complexity amplifies variance, and that regularisation is particularly beneficial when dimensionality is high relative to sample

size.

C. Jackknife Stability Analysis

Present:
Maximum jackknife SE per model
Possibly a plot of coefficient SE distribution
Highlight:
Multinomial MLE extreme instability
Ridge reduces variability significantly
Binary already stable
This is where your variance story becomes visible.

TABLE VI
MAXIMUM JACKKNIFE STANDARD ERROR BY MODEL

Model	Max Jackknife SE	Associated Coefficient
Binary MLE	0.571	cp_2.0
Binary Ridge	0.198	cp_2.0
Multinomial MLE	4.058	const (Class 4)
Multinomial Ridge	0.461	cp_2.0 (Class 3)

D. Jackknife Stability Analysis

Table VI reports the maximum jackknife standard error observed under each model. The binary MLE exhibits moderate variability, with a maximum coefficient SE of 0.571. Ridge regularisation reduces this to 0.198, indicating modest variance reduction in an already stable setting.

TABLE VII
SELECTED MULTINOMIAL JACKKNIFE SE (MLE VS RIDGE)

Predictor	Class	MLE SE	Ridge SE
const	4	4.058	0.322
cp_3.0	4	3.502	0.389
cp_4.0	4	3.485	0.333
cp_2.0	3	1.113	0.461

In contrast, the multinomial MLE displays extreme instability. The largest jackknife standard error exceeds 4.0 for the intercept of class 4, and several class-specific predictors exhibit SE values above 3.0 (Table VII). These values reflect severe finite-sample instability in rare outcome classes.

Ridge regularisation dramatically reduces these instabilities. For example, the class 4 intercept SE decreases from 4.058 to 0.322, and similarly large reductions are observed for several predictors. After regularisation, the maximum multinomial SE falls below 0.5, bringing it much closer to the binary model.

TABLE VIII
JACKKNIFE SE OF PREDICTED PROBABILITIES (MULTINOMIAL)

Class	MLE SE	Ridge SE	Ridge/MLE
0	0.0281	0.0268	0.95
1	0.0274	0.0250	0.91
2	0.0184	0.0136	0.74
3	0.0152	0.0132	0.87
4	0.0056	0.0045	0.80

At the level of predicted probabilities (Table VIII), ridge consistently reduces jackknife variability across all outcome classes, with reductions ranging from 5% to over 25%. The largest proportional improvements occur for the rarer classes.

These results provide direct evidence that increased model complexity substantially amplifies estimator variance in finite samples. Ridge regularisation mitigates this effect by shrinking coefficients, particularly in the high-dimensional multinomial setting. In contrast, the binary model is quite stable even without penalisation.

V. DISCUSSION

The results illustrate a consistent relationship between model complexity, estimator variance, and the benefits of regularisation in finite samples. Moving from a binary to a multinomial model explicitly increases flexibility by predicting multiple severity levels, but this comes at the cost of a substantially larger parameter space. This results in an increase in estimator variance because each non-reference class requires its own coefficient vector, while the available information per class is divided into more parts.

A key reason for the increased variance is class imbalance. Rare outcome categories contribute comparatively little information to the likelihood. This leads to large changes in parameter estimates, which is reflected empirically in the large jackknife standard errors observed for the multinomial MLE. The extreme coefficient magnitudes and instability for specific class–predictor combinations are consistent with quasi-separation, which is a known failure of MLE in sparse classification settings.

Ridge regularisation addresses this by shrinking coefficients toward zero and effectively stabilising optimisation. In practical terms, ridge reduces sensitivity to individual observations and prevents the model from fitting extreme class-specific patterns that are only weakly supported by the data. The cross-validated log-loss curves showcase how moderate regularisation improves generalisation by reducing variance, while overly strong regularisation eventually degrades performance as bias increases too much and the model loses its ability to capture genuine structure in the data.

VI. CONCLUSION

Despite the improvement from regularisation, the binary model achieved the best predictive performance overall. This is consistent with the idea that, under limited data, collapsing severity levels reduces modeling difficulty and yields a more robust estimator. The binary model pools information across all diseased cases, increasing the effective sample size and reducing variance without requiring strong shrinkage. In this finite-sample regime, the predictive benefit of the multinomial model's added flexibility does not outweigh its increased variance.

These findings also suggest that the dataset may be unfit for fully unconstrained multinomial modeling without strong regularisation or additional data. In particular, the rarest severity class contains very few observations relative to the number

of parameters required to estimate class-specific effects. With a larger sample size, especially in the lowest sample class, the multinomial model could plausibly become more competitive, as increased information would reduce variance and make asymptotic approximations more reliable. Alternatively, approaches that share strength across classes especially ordinal models may be better suited to severity modelling particularly because the data had an ordinal scale, meaning that 4 was the most serious and 1 the least serious heart condition.

Finally, the conclusions should only apply to this specific dataset and modeling setup. The observed levels of instability and improvement from ridge depend on factors such as class balance, feature dimensionality, and correlation structure. The central conclusion of the research is that increasing outcome complexity increases estimator variance in finite samples, and regularisation becomes increasingly more effective and important as dimensionality rises relative to available information.

REFERENCES

[1] D. Dua and C. Graff, "UCI Machine Learning Repository: Heart Disease Dataset," University of California, Irvine, School of Information and Computer Sciences, 2019. [Online]. Available: https://archive.ics.uci.edu/ml/datasets/heart+disease

VII. APPENDIX

A.

TABLE IX
DISTRIBUTION OF DATA ORIGIN

Origin	Count	Percent (%)
CLE	303	40.9
HUN	261	35.3
VA	130	17.6
SW	46	6.2

B.

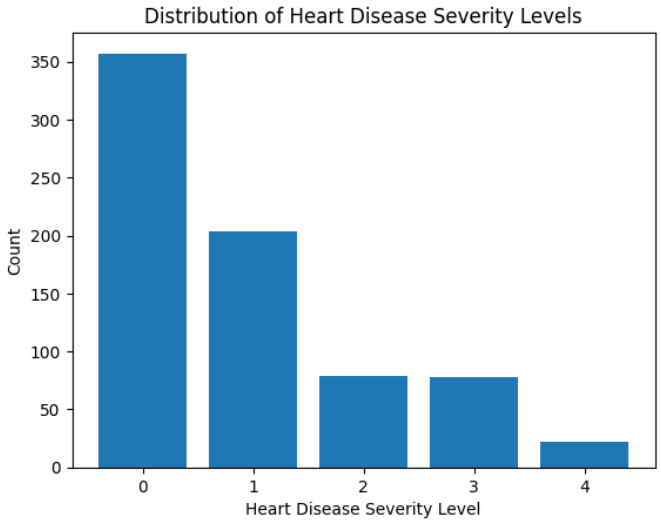


Fig. 2. Distribution of heart disease severity levels in the dataset.

C.

D.

TABLE X
JACKKNIFE STANDARD ERRORS FOR NON-PENALISED MULTINOMIAL LOGISTIC REGRESSION (MLE)

Predictor	Class 0	Class 1	Class 2	Class 3	Class 4
const	1.139	1.158	1.220	1.316	4.058
age	0.133	0.126	0.152	0.162	0.347
trestbps	0.126	0.098	0.105	0.142	0.242
chol	0.118	0.100	0.116	0.119	0.227
thalach	0.132	0.118	0.154	0.167	0.335
oldpeak	0.127	0.097	0.114	0.135	0.223
sex_1.0	0.381	0.380	0.492	0.464	1.256
cp_2.0	0.615	0.649	1.076	1.113	0.793
cp_3.0	1.019	1.041	1.054	1.209	3.502
cp_4.0	1.020	1.029	1.018	1.196	3.485
fbs_1.0	0.306	0.268	0.331	0.302	0.643
restecg_1.0	0.330	0.264	0.312	0.323	0.672
restecg_2.0	0.256	0.233	0.286	0.307	0.519
exang_1.0	0.257	0.229	0.277	0.297	0.590

TABLE XI
JACKKNIFE STANDARD ERRORS FOR RIDGE-REGULARISED MULTINOMIAL LOGISTIC REGRESSION

Predictor	Class 0	Class 1	Class 2	Class 3	Class 4
const	0.286	0.272	0.290	0.268	0.322
age	0.109	0.105	0.134	0.142	0.249
trestbps	0.110	0.087	0.095	0.134	0.182
chol	0.105	0.091	0.110	0.110	0.185
thalach	0.111	0.100	0.137	0.148	0.241
oldpeak	0.115	0.089	0.108	0.126	0.187
sex_1.0	0.194	0.203	0.282	0.279	0.330
cp_2.0	0.304	0.321	0.401	0.461	0.119
cp_3.0	0.298	0.300	0.320	0.301	0.389
cp_4.0	0.291	0.263	0.266	0.267	0.333
fbs_1.0	0.243	0.220	0.272	0.255	0.378
restecg_1.0	0.257	0.216	0.257	0.260	0.399
restecg_2.0	0.207	0.195	0.238	0.253	0.320
exang_1.0	0.206	0.188	0.237	0.248	0.391